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Freiman's Theorem - Proof and Best Bounds

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1 Introduction

In this report, we will look at the statement and motivation for Freiman's Theorem, a result in Additive Combinatorics, which states that any subset of integers, with a bounded doubling constant, must be contained in a General Arithmetic Progression (GAP), with bounded size and dimension. We will then look at some important tools required to prove this statement and then bring them all together to see the proof of the theorem. Finally, we will look at the various bounds that have been obtained on the size and dimension of the corresponding GAP, and look into how these were obtained.

2 Preliminaries

Definition 1. A **Generalized Arithmetic Progression (GAP)** of **dimension** d is a set of the form:

$$\{x_0 + l_1x_1 + \dots + l_dx_d \mid 0 \leq l_1 < L_1, \dots, 0 \leq l_d < L_d \in \mathbb{Z}\}$$

where $x_0, x_1, \dots, x_d \in \mathbb{Z}$ and $L_1, L_2, \dots, L_d \in \mathbb{N}$.

- **Size** of a GAP: $L_1L_2\dots L_d$.
- If there are no coincidences of elements, then the GAP is called **proper**.

Definition 2. The **doubling constant** of a finite subset, A , of an abelian group is the ratio $\frac{|A+A|}{|A|}$

We have seen that:

- APs have doubling constant at most 2.
- Proper GAPs have doubling constant at most 2^d .
- Any subset of an AP has a bounded doubling constant:
If B is a finite arithmetic progression and $A \subseteq B$ has $|A| \geq C|B|$, then $|A+A| \leq |B+B| \leq 2|B| \leq \frac{2}{C}|A| \implies A$ has doubling constant at most $\frac{2}{C}$.
- The same property holds for GAPs as well.
- Now we ask the converse question - if a set has bounded doubling constant, must it be contained in a GAP? - Freiman's Theorem answers this question!!

3 Freiman's Theorem

Theorem 3. If $A \subseteq \mathbb{Z}$ is a finite set and $|A+A| \leq K|A|$, then A is contained in a GAP of dimension at most $d(K)$ and size at most $f(K)|A|$, where $d(K)$ and $f(K)$ are constants depending only on K .

- The conclusion of the theorem can be obtained for a proper GAP as well. It comes at the cost of increasing $d(K)$ and $f(K)$. (Tao and Vu, 2006)

Necessary tools for the proof of Freiman's Theorem

- Plünnecke-Ruzsa Inequality & Ruzsa Covering Lemma

- Freiman Homomorphisms and Isomorphisms
- Ruzsa Modeling Lemma
- Bohr sets and Bogolyubov's Lemma
- Geometry of Numbers

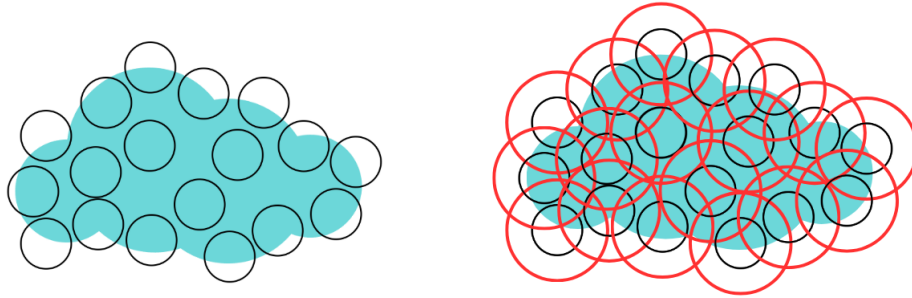
3.1 Plünnecke-Ruzsa Inequality & Ruzsa Covering Lemma

Theorem 4 (Plünnecke-Ruzsa Inequality). *If A is a finite subset of an abelian group and $|A + A| \leq K|A|$, then $|mA - nA| \leq K^{m+n}|A|$*

This inequality states that if $A \subset \mathbb{Z}$ has bounded doubling constant, then iterated sums and differences of A also have bounded doubling constant.

Theorem 5 (Ruzsa Covering Lemma). *Let X and B be subsets of an abelian group. If $|X + B| \leq K|B|$, then there exists a subset $T \subset X$ with $|T| \leq K$ such that $X \subset T + B - B$.*

This theorem can be intuitively thought of in terms of covering X with copies of $B - B$. It states that if it appears like $X + B$ can be covered by K many copies of B (in terms of size alone), then at most K copies of $B - B$, which corresponds to a ball of double radius of B (in a Euclidean space), can cover X . In the diagrams, the green cloud represents X , the black balls represent copies of B , and the red ones represent copies of $B - B$.



The proof of Ruzsa Covering Lemma is done by taking T , a maximal packing of disjoint copies of balls, B which have some intersection with X . We then construct $T + B - B$ using that maximal packing. Here, the first figure corresponds to the maximal packing, while the second one, with red balls represents $T + B - B$, and shows that it does indeed cover X !

3.2 Freiman Homomorphisms and Isomorphisms

We are familiar with group homomorphisms and know that group homomorphisms preserve complete additive structure. Now we are going to look at Freiman s -homomorphisms which preserve partial additive structure.

Definition 6. *Let A, B be subsets in (possibly different) abelian groups. We say that $\phi : A \rightarrow B$ is a **Freiman s -homomorphism** (or a Freiman homomorphism of order s), if*

$$\phi(a_1) + \dots + \phi(a_s) = \phi(a'_1) + \dots + \phi(a'_s)$$

whenever $a_1, \dots, a_s, a'_1, \dots, a'_s \in A$ satisfy

$$a_1 + \dots + a_s = a'_1 + \dots + a'_s$$

Recall that for a map to be a group isomorphism, it is sufficient for it to be a bijective group homomorphism. But here, since the condition for a Freiman s -homomorphism is weaker, we need an additional condition to define a Freiman s -isomorphism, which goes as follows.

Definition 7. If $\phi : A \rightarrow B$ is a bijection, and both ϕ and ϕ^{-1} are Freiman s -homomorphisms, then ϕ is said to be a **Freiman s -isomorphism**.

Examples:

1. Any group homomorphism is a Freiman s -homomorphism for any order, s .
2. If A is a group without any additive structure, (for example, $\{1, 10, 100, 1000\}$), and $B = \mathbb{Z}$, then any map, $\phi : A \rightarrow B$ is a Freiman 2-homomorphism.
3. Any $\phi : \{0, 1\} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a Freiman 2-homomorphism, and is bijective, but is not a Freiman 2-isomorphism. (Consider $\{0, 1\}$ as a subset of $\mathbb{Z}$ here)
4. In general, $\text{mod } N : \mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$ is a Freiman homomorphism of every order, but not a Freiman isomorphism.

Proposition 8. If $A \subset \mathbb{Z}$ has diameter smaller than N/s , then $(\text{mod } N)$ maps A Freiman s -isomorphically to its image.

Proof. It is sufficient to show that ϕ^{-1} is a Freiman s -homomorphism.

Consider $\phi : A \rightarrow \mathbb{Z}/N\mathbb{Z}$, where $\phi(x) = x \text{ mod } N$.

Let $a_1, \dots, a_s, a'_1, \dots, a'_s \in A$. Let $\phi(a_1) + \dots + \phi(a_s) = \phi(a'_1) + \dots + \phi(a'_s)$,

To show ϕ is a Freiman s -isomorphism, we now need to show that $a_1 + \dots + a_s = a'_1 + \dots + a'_s$.

$\phi(a_1) + \dots + \phi(a_s) = \phi(a'_1) + \dots + \phi(a'_s) \implies (a_1 + \dots + a_s) - (a'_1 + \dots + a'_s) = 0 \text{ mod } N$

Since the diameter of A is less than N/s , $|a_i - a'_i| < N/s$ for $1 \leq i \leq s$

$\implies |a_1 + \dots + a_s - a'_1 - \dots - a'_s| < N \implies (a_1 + \dots + a_s - a'_1 - \dots - a'_s) = 0 \quad \square$

Remarks:

- If two sets are Freiman 2-isomorphic, then APs are preserved between the two sets.
- If a subset B is Freiman 8-isomorphic to A' , then $2B - 2B$ is Freiman 2-isomorphic to $2A' - 2A'$. (Follows from the definition)

3.3 Ruzsa Modeling Lemma

Theorem 9. Let $A \subset \mathbb{Z}$, $s \geq 2$, and N be a positive integer such that $N \geq |sA - sA|$. Then there exists $A' \subset A$ with $|A'| \geq |A|/s$ such that A' is Freiman s -isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.

Idea of Proof.

- Let $q > \max(sA - sA)$ be a prime. For every $\lambda \in [q - 1]$, consider

$$\phi : \mathbb{Z} \xrightarrow{\text{mod } q} \mathbb{Z}/q\mathbb{Z} \xrightarrow{\cdot \lambda} \mathbb{Z}/q\mathbb{Z} \rightarrow [q].$$

For all λ , there exists an interval, $I_\lambda \subset [q]$ of length less than q/s such that $A_\lambda = \{a \in A : \phi(a) \in I_\lambda\}$ has more than $|A|/s$ elements. ϕ , when restricted to this A_λ is a Freiman s -homomorphism.

- Now consider $\psi : \mathbb{Z} \xrightarrow{\phi} [q] \xrightarrow{\text{mod } N} \mathbb{Z}/N\mathbb{Z}$

By some additional arguments, we can show that there is some λ such that ψ maps A_λ s -isomorphically to its image. We simply take $A' = A_\lambda$.

Corollary 10. *If $A \subset \mathbb{Z}$ with $|A + A| \leq K|A|$, then there exists a prime $N \leq 2K^{16}|A|$ and some $A' \subset A$ with $|A'| \geq |A|/8$ such that A' is Freiman 8-isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.*

Proof. $|8A - 8A| \leq K^{16}|A|$ (by Plünnecke Ruzsa Inequality).

Choose a prime, N , such that $K^{16}|A| \leq N < 2K^{16}|A|$. (by Bertrand's postulate)

Apply modeling lemma with $s = 8$ to get the required result. $\square$

3.4 Bohr sets and Bogolyubov's Lemma

Definition 11. Suppose $R \subset \mathbb{Z}/N\mathbb{Z}$. Define $\text{Bohr}(R, \epsilon) = \{x \in \mathbb{Z}/N\mathbb{Z} : \|\frac{rx}{N}\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon, \text{ for all } r \in R\}$. $|R|$ is the dimension of the set, denoted by d , and ϵ is called the width of the set.

Harald Bohr (1887-1951)



Harald Bohr, was a prominent mathematician, well-known for his work in Analysis, and for founding the field of almost periodic functions. The **Bohr set** in Additive Combinatorics is connected to this famous mathematician, who is also the brother of the physicist, Niels Bohr. Harald also worked with Hardy, Littlewood, Mollerup and Landau, among others. He even learned measure theory from Lebesgue himself!

Theorem 12 (Bogolyubov's Lemma). *If $A \subset \mathbb{Z}/N\mathbb{Z}$ and $|A| = \alpha N$, then $2A - 2A$ contains some Bohr set, $\text{Bohr}(R, 1/4)$, with $|R| < 1/\alpha^2$.*

Proof. Recall that $f \hat{*} g = \hat{f} \hat{g}$, and $1_{A+B} = 1_A * 1_B$, which implies that $1_A \hat{*} 1_B = \hat{1}_A \hat{1}_B$.

Using this iteratively, we get $1_{2A-2A} = (\hat{1}_A)^2 (\hat{1}_{-A})^2$.

$$\begin{aligned} \hat{1}_{-A}(x) &= \frac{1}{N} \sum_{r \in \mathbb{Z}/N\mathbb{Z}} \hat{1}_{-A}(r) e^{-2\pi r i x} = \frac{1}{N} \sum_{r \in -A} e^{-2\pi r i x} = \overline{\frac{1}{N} \sum_{r \in A} e^{-2\pi r i x}} = \overline{\hat{1}_A(x)} \\ \implies 1_{2A-2A}(r) &= |\hat{1}_A(r)|^4 \end{aligned}$$

And since 1_{2A-2A} is a real function, we get $1_{2A-2A}(x) = \sum_{r \in \mathbb{Z}/N\mathbb{Z}} |\hat{1}_A(r)|^4 \cos(\frac{2\pi r x}{N})$

Now we define R as: $\{r \in \mathbb{Z}/N\mathbb{Z} \setminus \{0\} \mid |\hat{1}_A(r)| > \alpha^{1.5}\}$. Let $x \in \text{Bohr}(R, 1/4)$.

From the definition of Bohr sets, $-\frac{\pi}{2} < \frac{2\pi r x}{N} < \frac{\pi}{2} \implies \cos(\frac{2\pi r x}{N}) > 0$ for all $r \in R$.

By Plancherel's identity, $\sum_{r \in \mathbb{Z}/N\mathbb{Z}} |\hat{f}(r)|^2 = \frac{1}{N} \sum_{r \in \mathbb{Z}/N\mathbb{Z}} |f(r)|^2$.

$$\implies \sum_{r \in \mathbb{Z}/N\mathbb{Z}} |\hat{1}_A(r)|^2 = \frac{1}{N} \sum_{x \in \mathbb{Z}/N\mathbb{Z}} |1_A(x)|^2 = \alpha$$

$$\implies \alpha \geq \sum_{r \in A} |\hat{1}_A(r)|^2 > \alpha^3 |R| \implies 1/\alpha^2 > |R|$$

If we show that $x \in 2A - 2A$, then we are done. So, we look at $1_{2A-2A}(x)$.

$$\begin{aligned} 1_{2A-2A}(x) &= \sum_{r \in \mathbb{Z}/N\mathbb{Z}} |\hat{1}_A(r)|^4 \cos(\frac{2\pi r x}{N}) \\ &= |\hat{1}_A(0)|^4 + \sum_{r \in R} |\hat{1}_A(r)|^4 \cos(\frac{2\pi r x}{N}) + \sum_{r \in (\mathbb{Z}/N\mathbb{Z}) \setminus (R \cup \{0\})} |\hat{1}_A(r)|^4 \cos(\frac{2\pi r x}{N}) > 0 \end{aligned}$$

Hence, $x \in 2A - 2A$, which concludes the proof. $\square$

What are Approximate Groups?

A subset, A , of a group, G is defined to be a **k -approximate group** of G if $A^{-1} = A$, $1 \in A$ and $\exists X \subset G$ with $|X| \leq K$ such that $A^2 \subset XA$. This was introduced by **Terence Tao**, who also showed that the notion of an approximate group is equivalent to small doubling!

Approximate groups are also closely connected to Bohr sets and Freiman Homomorphisms, with every Bohr set being an approximate group, and all approximate groups being stable under Freiman homomorphisms: given a k -approximate group, any Freiman 2-homomorphism satisfying certain basic properties, will map A to another k -approximate group.

3.5 Geometry of Numbers

Definition 13. A **lattice** in $\mathbb{R}^d$ is a set given by $\Lambda = \mathbb{Z}v_1 \oplus \dots \oplus \mathbb{Z}v_d$, where $v_1, \dots, v_d \in \mathbb{R}^d$ are linearly independent vectors.

Definition 14. The **determinant** $\det(\Lambda)$ of a lattice $\Lambda = \mathbb{Z}v_1 \oplus \dots \oplus \mathbb{Z}v_d$ is the absolute value of the determinant of the matrix with $v_1, \dots, v_d$ as columns.

Remark: The determinant of a lattice is also equal to the volume of the fundamental parallelepiped.

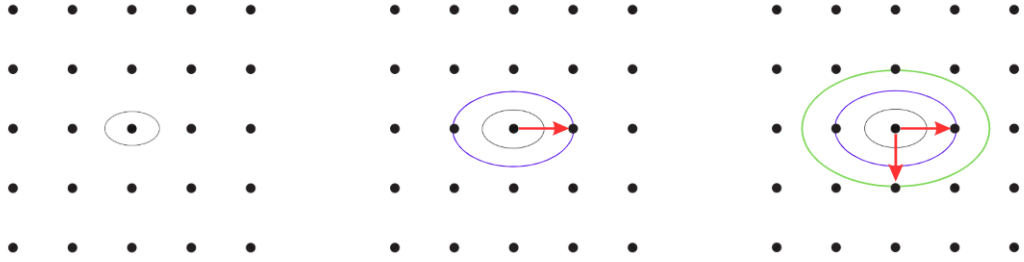
Definition 15. Given a centrally symmetric convex body $K \subset \mathbb{R}^d$ (centrally symmetric: $x \in K \iff -x \in K$), define the i^{th} successive minimum of K with respect to a lattice Λ as

$$\lambda_i = \inf\{\lambda \geq 0 : \dim(\text{span}(\lambda K \cap \Lambda)) \geq i\}$$

for $1 \leq i \leq d$. Equivalently, λ_i is the minimum λ such that λK contains i linearly independent lattice vectors from Λ . A directional basis of K with respect to Λ is a basis $b_1, \dots, b_d$ of $\mathbb{R}^d$ such that $b_i \in \lambda_i K$ for each $i = 1, \dots, d$.

What does λ_i represent?

We can visualise λ_i , as the factor by which K needs to be multiplied/scaled up to, such that the span of the intersection of $\lambda_i K$ and the lattice has dimension at least i . Consider the following example:



Here consider the black ellipse (first diagram), to be K . It initially contains only one point of the lattice, and hence is 0-dimensional. When we scale it up to reach the blue ellipse (second diagram), then the span of the intersection becomes a 1-dimensional space. The factor by which K is scaled up to reach the blue one is λ_1 . Similarly, for the span of the intersection to become 2-dimensional, we need to scale the ellipse up by λ_2 . $\lambda_2 K$ corresponds to the green ellipse (third diagram). We can also get a directional basis, $\{b_1, b_2\}$, such that $b_i \in \lambda_i K$, $i = 1, 2$. The red arrows correspond to one such basis.

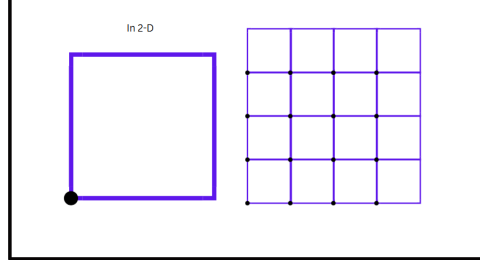
Theorem 16 (Minkowski's second theorem). Let $\Lambda \in \mathbb{R}^d$ be a lattice and K a centrally symmetric convex body. Let $\lambda_1 \leq \dots \leq \lambda_d$ be the successive minima of K with respect to Λ . Then,

$$\lambda_1 \dots \lambda_d \text{vol}(K) \leq 2^d \det(\Lambda)$$

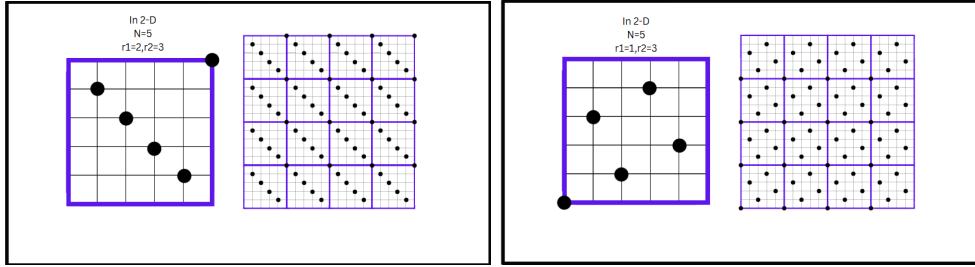
Theorem 17. Let N be a prime. Every Bohr set of dimension d and width $\epsilon \in (0, 1)$ in $\mathbb{Z}/N\mathbb{Z}$ contains a proper GAP with dimension at most d and size at least $(\epsilon/d)^d N$.

In the proof of this theorem, we want to connect Bohr sets to Lattices, and hence to GAPs. We will do this by considering a lattice that is constructed using the elements of a Bohr set, in such a way that a GAP in the lattice (which corresponds to a subset of the lattice points) is in the Bohr set.

Consider a 2-D lattice, with $v_1 = (1, 0)$ and $v_2 = (0, 1)$. We can construct it by taking a unit square and replicating it infinitely in both directions as follows: (the first square represents the unit square, and the diagram on the right is a representation of the extension - continuing with that extension will give the complete lattice)



Let $R = \{r_1, \dots, r_d\}$, $v = (\frac{r_1}{N}, \dots, \frac{r_d}{N})$. In the proof, we take the lattice to be $\Lambda \subset \mathbb{R}^d$ consisting of all points in $\mathbb{R}^d$ that are congruent mod 1 to some integer multiple of v . We can construct this lattice by similarly constructing the unit square corresponding to it and then replicating that infinitely in both directions. Let us look at a few examples to understand how that works:



- In the first figure, we consider $N = 5$, $r_1 = 2$ and $r_2 = 3$. Here $v = (\frac{2}{5}, \frac{3}{5})$. Hence, we first divide the unit lattice into 25 parts, with 5 rows and 5 columns. We now mark all the points $k(\frac{2}{5}, \frac{3}{5}) \bmod 1$, where $k = 1, 2, 3, 4, 5$. Since mod causes the points to wrap around, for k values after $N = 5$, the points repeat. So we need to consider only $k = 1, 2, \dots, N$. Now we can get the lattice by simply replicating this unit square. Here, again the diagram on the right of the unit square is a representation of the extension.
- The second figure does the same for $N = 5$, $r_1 = 1$, and $r_2 = 3$.
- Observe that the representation of a lattice (in terms of the values of N, r_1 and r_2) need not necessarily be unique. For example, consider the case with $N = 5$, $r_1 = 1$ and $r_2 = 4$. The lattice obtained in this case will be the same as the case of $N = 5$, $r_1 = 2$ and $r_2 = 3$.

Now let us look at the proof of the theorem:

Proof. Let $R = \{r_1, \dots, r_d\}$, $v = (\frac{r_1}{N}, \dots, \frac{r_d}{N})$. Consider the lattice $\Lambda \subset \mathbb{R}^d$ consisting of all points in $\mathbb{R}^d$ that are congruent mod 1 to some integer multiple of v .

Note that $\det(\Lambda) = 1/N$. Consider $K = [-\epsilon, \epsilon]^d$.

Let $\lambda_1, \dots, \lambda_d$ be the successive minima of K with respect to Λ .

Let $b_1, \dots, b_d$ be the directional basis.

By construction, $\|b_j\|_\infty \leq \lambda_j \epsilon$ for all $1 \leq j \leq d$.

For each j , let $L_j = \lceil 1/(\lambda_j d) \rceil$. $0 \leq l_j < L_j \implies \|l_j b_j\|_\infty < \frac{\epsilon}{d}$.

Hence, $\|l_1 b_1 + \dots + l_d b_d\|_\infty \leq \epsilon$.

Since each b_j is $x_j v$ plus a vector with integer coordinates for some $0 \leq x_j < N$,

we have $\left\| \frac{(l_1 x_1 + \dots + l_d x_d) r_i}{N} \right\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon$ for all i .

Hence the GAP, $\{l_1 x_1 + \dots + l_d x_d : 0 \leq l_i \leq L_i \text{ for all } i\}$ is in $\text{Bohr}(R, \epsilon)$.

We can show that it's size is large using Minkowski's theorem. We also need to show that it is proper, which we can do by similarly bounding the norm of $b = (l_1 - l'_1)b_1 + \dots + (l_d - l'_d)b_d$ by ϵ , forcing b to be 0.

□

4 Proof of Freiman's Theorem

Now, let us look at the proof of Freiman's theorem. This part will use all of the results and ideas that we have looked at so far. The overall idea is as follows:

We start with a finite subset of integers, A . Using Ruzsa's Modelling Lemma, we get A' , a large subset of A , which is Freiman 8-isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$, B , where N is a prime. Since we have tools like Fourier analysis in $\mathbb{Z}/N\mathbb{Z}$, we use them to find a GAP inside $2B - 2B$, which we map to a GAP in $2A' - 2A'$ using a Freiman Isomorphism. We then use this GAP to cover the set A , which gives us a GAP that A is contained in, by the covering lemma. Given below are the details (which also give the respective bounds, $f(K)$ and $d(K)$):

Proof. ■ Start with a set A with $|A + A| \leq K|A|$. By the corollary of the Ruzsa Modeling Lemma, we map $A' \subset A$ (with $|A'| \geq |A|/8$) to $B \subset \mathbb{Z}/N\mathbb{Z}$, where N is a prime such that $N \leq 2K^{16}|A|$, and A' and B are 8-Freiman isomorphic.

- Take $\alpha = \frac{|B|}{N} = \frac{|A'|}{N} \geq \frac{|A|}{8N} \geq \frac{1}{16K^{16}}$.

Bogolyubov's Lemma $\implies 2B - 2B$ contains a Bohr set, $\text{Bohr}(R, 1/4)$, where $|R| < 1/\alpha^2 \leq 256K^{32}$

- By the geometry of numbers, there is a proper GAP inside this Bohr set, with $d < 256K^{32}$ and size $\geq (\frac{\epsilon}{d})^d N = (4d)^{-d} N$.
- This GAP is mapped to a proper GAP, say Q , in $2A' - 2A'$ (by properties of Freiman's isomorphisms), with same dimension and size.

- $Q \subset 2A - 2A \implies Q + A \subset 3A - 2A$.

By Plünnecke-Ruzsa inequality, $|Q + A| \leq |3A - 2A| \leq K^5|A|$

- $|Q| \geq (4d)^{-d} N$ and $N \geq |A'| \geq |A|/8$

$$\implies |Q + A| \leq K'|Q|, \text{ where } K' = 8(4d)^d K^5 = 8K^5(1024K^{32})^{256K^{32}} = e^{K^{O(1)}}$$

- By Ruzsa Covering Lemma, $\exists X \subset A$, with $|X| \leq K'$, such that $A \subset X + Q - Q$.
- X is contained in a GAP with dimension $|X|$ with length 2 in each direction, and $Q - Q$ is a GAP with same dimension, d , as Q .

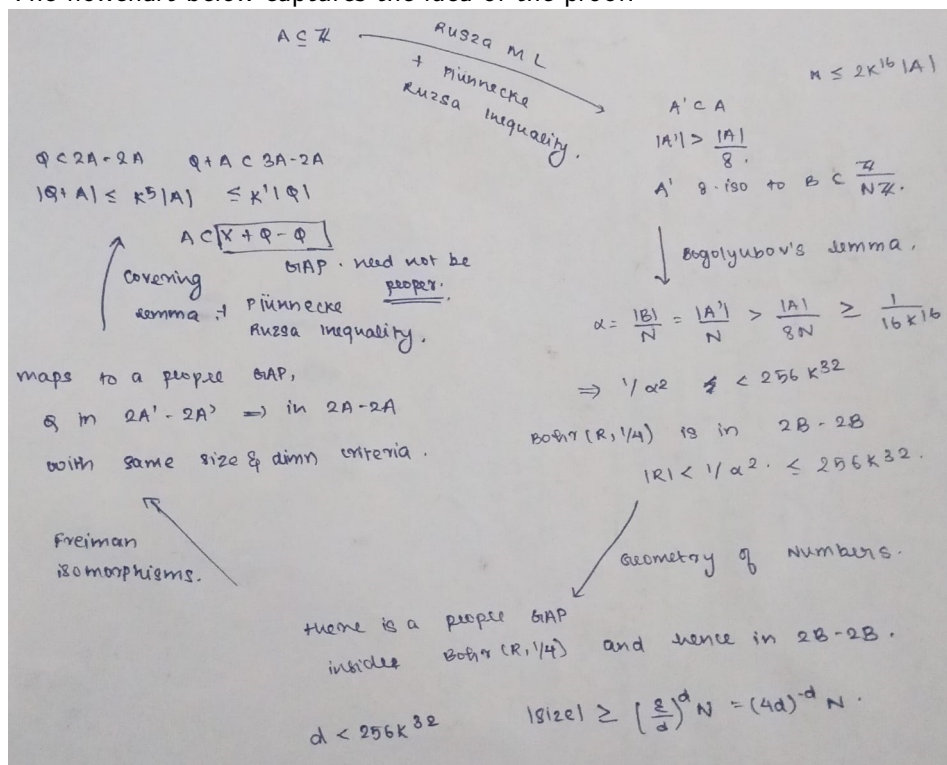
Hence, $X + Q - Q$ is contained in a GAP of

$$\text{dimension} \leq |X| + d \leq K' + d \leq d(K),$$

$$\text{and size} \leq 2^{|X|+d}|Q| \leq 2^{K'+d}|2A - 2A| \leq 2^{K'+d}K^4|A| \leq f(K)|A|$$

□

The flowchart below captures the idea of the proof:



5 Best bounds

- With the above proof (Ruzsa), we get the bounds: $d(K) = e^{K^{O(1)}}$ and $f(K) = e^{e^{K^{O(1)}}}$.
- Chang(2002) showed that by modifying the covering lemma part of the proof, we can get polynomial bounds $d(K) = K^{O(1)}$ and $f(K) = e^{K^{O(1)}}$.
- Sanders(2012) proved the bounds, $d(K) = K(\log K)^{O(1)}$ and $f(K) = e^{K(\log(K))^{O(1)}}$, which are the best bounds obtained so far.
- Conjectured bounds are $d(K) = O(K)$ and $f(K) = 2^{O(K)}$. (Open Problem!)

Idea of Chang's proof:

In Chang's proof, he improves the last two steps of the proof that we saw above. The covering lemma is wasteful in the last step, because of multiple overlaps between the translates of $Q - Q$. Chang improvised on this to get his better bound. His method takes the idea in the proof of Ruzsa's covering lemma and builds on that. Instead of taking one maximal subset and using translates of that to cover A , he proceeds iteratively, using a maximal subset at each step. This reduces overlaps, and is a more efficient way of covering A .

Idea of Sanders' proof:

Sanders' proof also proceeds along the same lines as Ruzsa's proof. He initially proves the **Bogolyubov–Ruzsa lemma for abelian groups**, which says that there is some $S \subset \mathbb{Z}$, such that there is a GAP contained in $A - A + S - S$, with appropriate bounds. He uses fourier analysis, Bohr sets, and lattices as a part of this proof. From here, he then uses Chang's covering Lemma to obtain a GAP that A is in, with the required bounds.

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